

JEE MAIN REVISION MASTER: HYPERBOLA

Subject: Mathematics | Target Scope: JEE Main Previous Years | Empirical Base: 3 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Moderate	Hyperbola: Eccentricity, Foci & Rectangular Hyperbola	3 PYQs	Conics

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Hyperbola: Eccentricity, Foci & Rectangular Hyperbola	$e = \sqrt{1 + \frac{b^2}{a^2}} > 1, \quad \text{Rectangular Hyperbola: } e = \sqrt{2},$ $ SP - S'P = 2a$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Hyperbola: Eccentricity, Foci & Rectangular Hyperbola (Moderate Yield — 3 PYQs)

Core Principle: Standard hyperbola $(x^2/a^2) - (y^2/b^2) = 1$: Eccentricity $e = \sqrt{1 + b^2/a^2} > 1$. Foci $(\pm ae, 0)$, Directrices $x = \pm a/e$, Length of latus rectum $= (2b^2/a)$. Rectangular (equilateral) hyperbola $x^2 - y^2 = a^2$ or $xy = c^2$ has eccentricity $e = \sqrt{2}$ and mutually perpendicular asymptotes.

■ **Caution / JEE Pitfall:** Eccentricity of a rectangular hyperbola is ALWAYS $\sqrt{2}$, independent of the value of a or c !

■ **Speed Trick:** Relation between eccentricities of a hyperbola (e_1) and its conjugate hyperbola (e_2): $(1/e_1^2) + (1/e_2^2) = 1$.

Representative Source PYQs:

- [2026 Shift] Q#1 (MCQ): "In the figure Statement-I: When $\alpha > \beta \geq 0$, the section is hyperbola Statement-II: When $\beta > 90^\circ$, the section is ellip..." [Ans: N/A]
- [2023 Shift] Q#2 (MCQ): "The distance between the foci of a hyperbola is 16 and its eccentricity is $\sqrt{2}$. Its equation is..." [Ans: N/A]
- [2018 Shift] Q#3 (MCQ): "The distance between the foci of a hyperbola is 16 and its eccentricity is $\sqrt{2}$. Its equation is..." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2026 (Q#1)

In the figure Statement-I: When $\alpha > \beta \geq 0$, the section is hyperbola Statement-II: When $\beta > 90^\circ$, the section is ellipse Which of the following is correct?

- (A) Statement I is true, Statement II is false
- (B) Statement I is false, Statement II is true
- (C) Both the Statements are true
- (D) Both the Statements are false

Correct Answer: None

Solution Analysis: Let α be the semi-vertical angle of the cone, and let β be the angle between the cutting plane and the axis of the cone. The conic section depends on the comparison between α and β . If $0 \leq \beta < \alpha$, the plane cuts both parts, or nappes, of the cone. The section formed is a hyperbola. If $\beta = \alpha$, the plane is parallel to a generator of the cone. The section formed is a parabola. If $\alpha < \beta < 90^\circ$, the plane cuts only one nappe of the cone. The section formed is an ellipse. If $\beta = 90^\circ$, the plane is perpendicular to the axis of t